

Baptiste PRAS

Master's student in Artificial Intelligence

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EDUCATION

Master's degree in Artificial Intelligence (PhD Track) <i>University Paris-Saclay</i>	Sep 2025 – Present Orsay, France
Magistère in Computer Science (Honors Research Program) <i>University Paris-Saclay</i> Graduated with Honors	Sep 2024 – Aug 2025 Orsay, France
Double Bachelor's Degree in Computer Science and Mathematics <i>University Paris-Saclay</i>	Sep 2022 – Aug 2024 Orsay, France
Intensive English Language Program <i>EF New York</i> Achieved C2 Proficiency	Sep 2019 – May 2020 New York, USA

PUBLICATIONS

- Point-Based Counting of Cereals and Legumes in Intercropped Fields**
Junior Conference on Data Science & Engineering (Poster), 2026
- Fine-Grained Mention-Level Analysis of Biomedical Entity Linking Models**
Medical Informatics Europe (Full Paper), 2026
- Revisiting Class Imbalance Ratios in Imbalanced Learning**
Junior Conference on Data Science & Engineering (Full Paper), 2025

PROFESSIONAL EXPERIENCE

- Research Internship** May 2026 – Aug 2026
Inria TAU Team - LISN Orsay, France
- Built an automated per-species counting pipeline for wheat and pea in intercropped plots from UAV and smartphone imagery. Benchmarked class-agnostic few-shot and crowd-counting models, engineering the full preprocessing pipeline: soil noise filtering, isotropic square tiling, and leaf-tip rather than whole-plant point annotations. Finetuned the point-query transformer PET with optimal input scaling and edge-border filtering, reaching 4.69% MAPE with 74.2% F1 on wheat tips and 7.83% MAPE with 86.2% F1 on pea.
- Contact: François Landes (LISN, Université Paris-Saclay)
- Supervised Research Project** Jan 2026 – Mar 2026
LISN - Laboratoire Interdisciplinaire des Sciences du Numérique Orsay, France
- Investigated Biomedical Entity Linking (BEL) strategies to normalize PICO entities (Population, Intervention, Comparison, Outcome) to the MeSH knowledge base. Developed and evaluated a hybrid normalization pipeline, comparing a custom rule-based script with a state-of-the-art graph-based entity linking model.
- Research Internship** May 2025 – Aug 2025
LISN - Laboratoire Interdisciplinaire des Sciences du Numérique Orsay, France
- Analyzed biomedical entity linking models on the BELB benchmark, focusing on generalization to rare or complex mentions. Developed quantitative and visual analyses of dataset characteristics (mention length, ambiguity, frequency) and their impact on prediction quality. Compared recent models, identified consistent weaknesses, and proposed improvements; published a paper in MIE.
- Contact: Nona Naderi (LISN, Université Paris-Saclay)
- Supervised Research Project** Jan 2025 – May 2025
LISN - Laboratoire Interdisciplinaire des Sciences du Numérique Orsay, France
- Studied the impact of class imbalance on classification tasks using a spherical Teacher-Student perceptron. Conducted experiments in Python (Scikit-Learn, NumPy, Matplotlib) with different noise levels, loss functions, and training methods (GD, Langevin dynamics). Demonstrated that the optimal imbalance ratio in training sets differs from 0.5 in class-imbalanced problems.

Generative AI Trainer

Jan 2025 – May 2026

Outlier / Alignerr

Remote

- Designed and refined prompts to enhance the performance of generative AI models. Reviewed and corrected AI-generated outputs to ensure accuracy and quality. Contributed to the continuous improvement of deep learning models through feedback and prompt optimization.

PROJECTS

Structure Detection in Fusion Plasma Simulations 🔗

- Developed a multi-stage detection pipeline featuring a YOLOv8 architecture optimized with a custom MLP post-filter for low-contrast frames, incorporating a pseudo-labeling strategy to expand a limited 30-image dataset and a feature extractor targeting intensity statistics and Sobel gradients.

Measuring Market Impact of Financial News 🔗

- Engineered a computationally efficient framework to transform unstructured financial news into (date, ticker, impact summary) event representations by implementing a hierarchical Map-Reduce summarization architecture with compact Transformers, ensuring numerical fidelity and entity grounding while validating results through a finance-oriented LLM-as-a-judge audit.

NBA MVP Prediction Model 🔗

- Collected and cleaned player and team statistics from publicly available online sources to build a structured dataset. Performed feature selection and engineering, experimenting with different attribute selection strategies. Using several predictive models implemented with Scikit-Learn, achieved an accuracy > 80% in predicting the actual MVP outcome.

Traffic Sign Recognition Model 🔗

- Developed a machine learning model to recognize traffic signs from pictures. Implemented preprocessing (normalization, resizing, feature extraction) and trained multiple supervised learning algorithms, reaching over 95% accuracy.

Dual Sudoku AI Agent 🔗

- Designed an artificial intelligence to solve and play Dual Sudoku by combining search algorithms and heuristic strategies. Implemented efficient state representation and evaluation functions to handle the game's combinatorial complexity.

Java-like Interpreter 🔗

- Built an interpreter supporting basic arithmetic, classes, methods, and Java-like type checking. Entirely coded in OCaml with Ocamllex and Menhir.

Other projects

- Developed different AI systems (Information Retrieval, CycleGAN), various algorithmic implementations (e.g., Graphs, Linked Trees, Quadrees) and 2D games (e.g., Air Hockey, Labyrinth solver, Colt Express, Worms) across Python, OCaml, Java, and C++.

SKILLS

Programming:

- Python (PyTorch, Scikit-Learn, NumPy, Matplotlib), Bash, C/C++, Java, OCaml, SQL

Artificial Intelligence:

- Deep Learning, Natural Language Processing, Computer Vision

Tools:

- LaTeX, Git, Linux, Slurm (Cluster Computing), Microsoft Office Suite

ACHIEVEMENTS

Winner of an AI competition by designing the best-performing Dual Sudoku agent

3x Prologon finalist & 1x Gradient finalist, national algorithmic programming contests

LANGUAGES

French

Native

Russian

Conversational

English

Bilingual Proficiency

TOEFL iBT: 108/120

TOEIC: 990/990